

Name _____

Teacher _____

**GOSFORD HIGH SCHOOL**

2012

HIGHER SCHOOL CERTIFICATE**ASSESSMENT TASK 3****MATHEMATICS – EXTENSION 1****Duration-** 60 minutes plus 5 minutes reading time**Special Instructions** - Tear off back page (multiple choice response/standard integrals)
- Each section will be collected separately

Section 1 Multiple choice	4 questions worth 1 mark each. (Answer this section on the multiple choice response sheet provided)	/4
Section 2 Inverse Functions	Answer this section on your own paper.	/9
Section 3 Inverse Trig Functions	Answer this section on your own paper.	/9
Section 4 General Solutions	Answer this section on your own paper.	/9
Section 5 Integration involving Trigonometric Functions	Answer this section on your own paper.	/9
TOTAL		/40

Section 1 Multiple Choice

(Total 4 marks)

Answer on the multiple choice response sheet provided. Each question is worth 1 mark.

Question 1

The range of $y = a + b \tan^{-1}(x - c)$ where a , b and c are positive real constants is

- A. $\frac{-\pi}{2} < y < \frac{\pi}{2}$
- B. $a - \frac{c\pi}{2} < y < a + \frac{c\pi}{2}$
- C. $c - \frac{b\pi}{2} < y < c + \frac{b\pi}{2}$
- D. $a - \frac{b\pi}{2} < y < a + \frac{b\pi}{2}$

Question 2

The curve given by $y = \sin^{-1}(2x)$ where $0 \leq x \leq \frac{1}{2}$, is rotated about the y-axis to form a solid of revolution. The volume of the solid may be found by evaluating

- A. $\frac{\pi}{4} \int_0^{\frac{\pi}{2}} (1 - \cos(2y)) dy$
- B. $\frac{\pi}{8} \int_0^{\frac{1}{2}} (1 - \cos(2y)) dy$
- C. $\frac{\pi}{8} \int_0^{\frac{\pi}{2}} (1 - \cos(2y)) dy$
- D. $\frac{1}{8} \int_0^{\frac{\pi}{2}} (1 - \cos(2y)) dy$

Question 3

Consider the function $f(x) = \frac{1}{1+x^2}$ where $x \geq 0$.

Which of the following best represents the inverse function for $f(x)$?

- A. $f^{-1}(x) = \frac{1}{1+y^2}$ for $y \geq 0$
- B. $f^{-1}(x) = \frac{1}{\sqrt{x}} - 1$ for $x \geq 0$
- C. $f^{-1}(x) = \pm \sqrt{\frac{1}{x} - 1}$ for $x > 0$
- D. $f^{-1}(x) = \sqrt{\frac{1}{x} - 1}$ for $x > 0$

Question 4

Which of the following is **NOT** true for the function $y = \sin^{-1} x$

- A. It is an odd function
- B. It is equivalent to $x = \sin y$ for all real y
- C. It has a domain of $-1 \leq x \leq 1$
- D. It is differentiable for the domain $-1 < x < 1$

End of multiple choice section

Section 2 Inverse Functions

(Total 9 marks)

- a. (i) If $f(x) = e^{x+2}$, find $f^{-1}(x)$ 2
- (ii) On the same axes, neatly sketch the graphs of $f(x)$ and $f^{-1}(x)$ 2
- b. (i) Explain why the function $f(x) = \sqrt{x-2}$ has an inverse function $f^{-1}(x)$ 1
- (ii) Write down the domain and range of $f(x)$ and $f^{-1}(x)$ 2
- c. Show that the following pairs of functions $f(x) = 2x - 1$ and $g(x) = \frac{x+1}{2}$ are inverses by showing that $f[g(x)] = g[f(x)] = x$ 2

Start a new page

Section 3 Inverse Trigonometry

(Total 9 marks)

- a. Determine the exact value of $\cos \left\{ 2 \sin^{-1} \left(\frac{12}{13} \right) \right\}$ 2
- b. Differentiate $y = \ln(\cos^{-1} x)$ 2
- c. (i) Write down the result for $\tan 2x$ in terms of $\tan x$ 1
- (ii) Using the fact that $\tan \frac{\pi}{4} = 1$ and considering the above result,
Show that $\tan \frac{\pi}{8} = \sqrt{2} - 1$ 2
- (iii) By consideration of the graph $y = \tan^{-1}(x - 1)$ and using the
result from part (ii), find the minimum value of a 2
(where a is a constant) for which $\tan^{-1}(x - 1) + a \tan \frac{\pi}{8} > 0$
for all x .

Start a new page

Section 4 General Solutions to Trigonometric Equations (Total 9 marks)

- a. Find the general solutions of $\tan \theta = 1$ 1
- b. Find the general solutions to $\sin^2 2x = \frac{3}{4}$ 2
- c. Find the general solutions to $3 \cos \theta - \sqrt{3} \sin \theta + 3 = 0$ 3
- d. Write down the general solutions to $\cos 2\theta = \sin \theta$ 3

Start a new page

Section 5 Integration involving Trigonometric Functions (Total 9 marks)

- a. Use the table of standard integrals to find the exact value of 2

$$\int_0^{\frac{\pi}{6}} \sec 4x \tan 4x \, dx$$

- b. Find $\int \cos^2 2x \, dx$ 3

- c. Find $\int \tan^2 2x \, dx$ 1

- d. Evaluate 3

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} \frac{x}{\sqrt{1-x^2}} \, dx \quad (\text{using the substitution } x = \cos \theta)$$

END OF EXAMINATION

Name: _____

Teacher: _____

Multiple-choice answer sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely, using a black pen.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct
↓

Start here → 1. A ☐ B ☐ C ☐ D ☐

2. A ☐ B ☐ C ☐ D ☐

3. A ☐ B ☐ C ☐ D ☐

4. A ☐ B ☐ C ☐ D ☐

FINAL ANSWERS TO R12 ext 1 task 3

2012

Section 1

$$Q1/ -\frac{\pi}{2} \leq \frac{y-a}{b} < \frac{\pi}{2}$$

$$a - \frac{b\pi}{2} < y < a + \frac{b\pi}{2} \quad (D)$$

$$Q2/ \begin{aligned} 2x &= \sin y \\ x &= \frac{1}{2} \sin y \\ x^2 &= \frac{1}{4} \sin^2 y \end{aligned}$$

$$\begin{aligned} \cos 2y &= 1 - 2\sin^2 y \\ \frac{1 - \cos 2y}{2} &= \sin^2 y \end{aligned}$$

$$\text{when } x=0$$

$$y=0$$

$$\text{when } x = \frac{1}{2}$$

$$y = \sin^{-1} 1 = \frac{\pi}{2}$$

$$\frac{\pi}{8} \int_0^{\frac{\pi}{2}} (1 - \cos 2y) dy$$

(C)

$$Q3/ x = \frac{1}{1+y^2}$$

$$1+y^2 = \frac{1}{x}$$

$$y^2 = \frac{1}{x} - 1$$

$$y = \pm \sqrt{\frac{1}{x} - 1}$$

$$\text{For } f^{-1}(x) \quad \begin{aligned} D: x > 0 \\ R: y \geq 0 \end{aligned}$$

$$\therefore y = \sqrt{\frac{1}{x} - 1} \text{ for } x > 0$$

(D)

Q4/ (B)

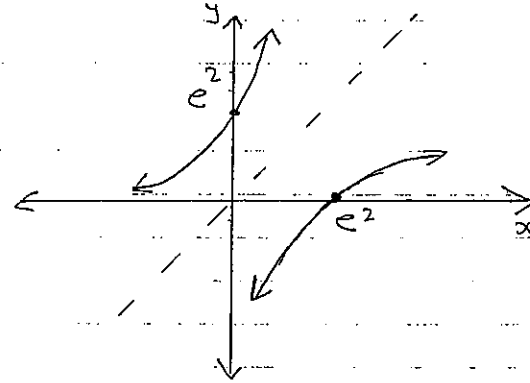
Section 2

$$(a) x = e^{y+2}$$

$$\ln x = y+2$$

$$\ln x - 2 = y$$

$$\therefore f^{-1}(x) = \ln x - 2$$



(b)(i) For $f(x)$, for every x value there is only one $f(x)$ value and for every $f(x)$ value there is only one x value.

$$(ii) \text{ For } f(x) \quad \begin{aligned} D: x \geq 2 \\ R: y \geq 0 \end{aligned}$$

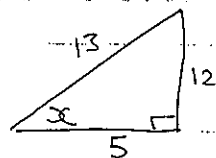
$$\text{For } f^{-1}(x) \quad \begin{aligned} D: x > 0 \\ R: y \geq -2 \end{aligned}$$

$$\begin{aligned} (c) f[g(x)] &= 2\left(\frac{x+1}{2}\right) - 1 \\ &= x+1-1 \\ &= x \end{aligned}$$

$$\begin{aligned}
 g[f(x)] &= \frac{(2x-1)+1}{2} \\
 &= \frac{2x}{2} \\
 &= x
 \end{aligned}$$

Section 3

(a) Let $x = \sin^{-1}\left(\frac{12}{13}\right)$



$$\begin{aligned}
 \therefore \cos 2x &= \cos^2 x - \sin^2 x \\
 &= \left(\frac{5}{13}\right)^2 - \left(\frac{12}{13}\right)^2 \\
 &= \frac{25}{169} - \frac{144}{169} \\
 &= \frac{-119}{169}
 \end{aligned}$$

(b) $y = \ln(\cos^{-1} x)$

$$\begin{aligned}
 y' &= \frac{-1}{\sqrt{1-x^2}} \div \cos^{-1} x \\
 &= \frac{-1}{\cos^{-1} x \sqrt{1-x^2}}
 \end{aligned}$$

(c) (i) $\tan 2x = \frac{2 \tan x}{1 - \tan^2 x}$

(ii) $\tan \frac{\pi}{4} = \frac{2 \tan \frac{\pi}{8}}{1 - \tan^2 \frac{\pi}{8}}$

$$1 - \tan^2 \frac{\pi}{8} = 2 \tan \frac{\pi}{8}$$

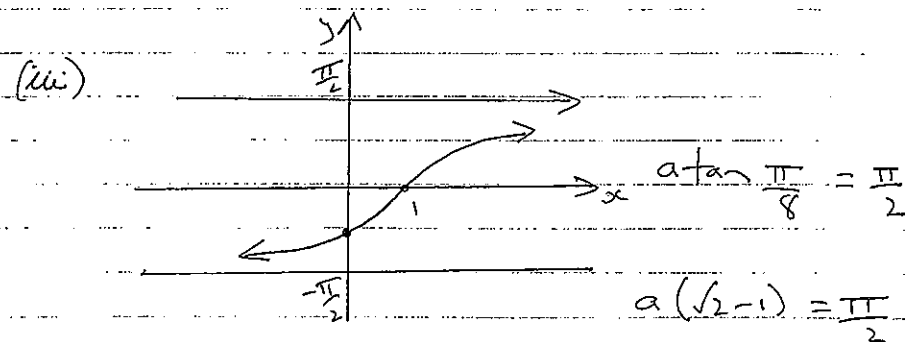
$$0 = \tan^2 \frac{\pi}{8} + 2 \tan \frac{\pi}{8} - 1$$

$$\begin{aligned}
 \tan \frac{\pi}{8} &= \frac{-2 \pm \sqrt{4 - 4 \times 1 \times -1}}{2} \\
 &= \frac{-2 \pm \sqrt{8}}{2}
 \end{aligned}$$

$$= \frac{-2 \pm 2\sqrt{2}}{2}$$

$$= -1 \pm \sqrt{2} \quad \text{but } \tan \frac{\pi}{8} > 0$$

$$\therefore \tan \frac{\pi}{8} = -1 + \sqrt{2}$$



$$a = \frac{\pi}{2\sqrt{2}-2}$$

Section 4

$$(a) \quad \Theta = n\pi + \tan^{-1} 1 \\ = n\pi + \frac{\pi}{4}$$

$$(b) \quad \sin 2x = \pm \frac{\sqrt{3}}{2} \\ 2x = n\pi + (-1)^n \sin^{-1} \pm \frac{\sqrt{3}}{2}$$

$$2x = n\pi + (-1)^n \left(\pm \frac{\pi}{3} \right)$$

$$x = \frac{n\pi}{2} + (-1)^n \left(\pm \frac{\pi}{6} \right)$$

$$(c) \quad 3 \cos \Theta - \sqrt{3} \sin \Theta = -3$$

$$R = \sqrt{9 + 3} \quad \tan \alpha = \frac{\sqrt{3}}{3} \\ = \sqrt{12} \\ = 2\sqrt{3} \quad \alpha = \frac{\pi}{6}$$

$$2\sqrt{3} \cos \left(\Theta + \frac{\pi}{6} \right) = -3$$

$$\cos \left(\Theta + \frac{\pi}{6} \right) = \frac{-3}{2\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \frac{-3\sqrt{3}}{6}$$

$$\Theta + \frac{\pi}{6} = 2n\pi \pm \frac{5\pi}{6}$$

$$= \frac{-\sqrt{3}}{2}$$

$$\Theta = 2n\pi \pm \frac{5\pi}{6} - \frac{\pi}{6}$$

$$\Theta = 2n\pi + \frac{2\pi}{3} \text{ OR}$$

$$\Theta = 2n\pi - \pi$$

$$(d) \quad \cos 2\Theta = \sin \Theta \\ 1 - 2\sin^2 \Theta = \sin \Theta \\ 0 = 2\sin^2 \Theta + \sin \Theta - 1$$

$$\begin{array}{ccc} 2\sin \Theta & & -1 \\ & \times & \\ \sin \Theta & & +1 \end{array}$$

$$(2\sin \Theta - 1)(\sin \Theta + 1) = 0 \\ \sin \Theta = \frac{1}{2} \text{ OR } \sin \Theta = -1$$

$$\Theta = n\pi + (-1)^n \cdot \frac{\pi}{6} \text{ OR } n\pi + (-1)^n \left(-\frac{\pi}{2} \right)$$

Section 5

$$(a) \quad \frac{1}{4} \left[\sec 4x \right]_0^{\frac{\pi}{6}} = \frac{1}{4} \left[\frac{1}{\cos \frac{2\pi}{3}} - \frac{1}{\cos 0} \right] \\ = \frac{1}{4} \times \left[-\frac{1}{2} - \frac{1}{1} \right] \\ = \frac{1}{4} \times [-2 - 1] \\ = -\frac{3}{4}$$

$$(b) \quad \cos 4x = \cos^2 2x - \sin^2 2x \\ = 2\cos^2 2x - 1$$

$$\cos 4x + 1 = 2\cos^2 2x$$

$$\cos^2 2x = \frac{1}{2} [\cos 4x + 1]$$

$$\begin{aligned}
 &= \frac{1}{2} \int \cos 4x + 1 \, dx \\
 &= \frac{1}{2} \left[\frac{1}{4} \sin 4x + x \right] + C \\
 &= \frac{\sin 4x}{8} + \frac{x}{2} + C
 \end{aligned}$$

$$(c) \int \sec^2 2x - 1 \, dx = \frac{1}{2} \tan 2x - x + C$$

$$\begin{aligned}
 (d) \quad x &= \cos \theta \\
 \frac{dx}{d\theta} &= -\sin \theta \\
 dx &= -\sin \theta \, d\theta
 \end{aligned}$$

$$\begin{aligned}
 \text{When } x &= \frac{1}{2} \\
 \theta &= \cos^{-1}\left(\frac{1}{2}\right) \\
 \theta &= \frac{\pi}{3}
 \end{aligned}$$

$$\begin{aligned}
 \text{When } x &= -\frac{1}{2} \\
 \theta &= \cos^{-1}\left(-\frac{1}{2}\right) \\
 \theta &= \frac{2\pi}{3}
 \end{aligned}$$

$$\int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \frac{-\sin \theta \cos \theta}{\sqrt{\sin^2 \theta}} \, d\theta$$

$$= - \int_{\frac{\pi}{3}}^{\frac{2\pi}{3}} \cos \theta \, d\theta$$

$$= \left[-\sin \theta \right]_{\frac{\pi}{3}}^{\frac{2\pi}{3}}$$

$$\begin{aligned}
 &= -\sin \frac{2\pi}{3} + \sin \frac{\pi}{3} \\
 &= -\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} \\
 &= 0
 \end{aligned}$$

NB If students recognise
 it is an odd
 function and give a
 bald answer of zero,
 they have ~~not~~ shown
 the substitution